

A 4.27-DAY NEPTUNE-MASS PLANET CANDIDATE AROUND THE K DWARF HD 297396 IN 18 YEARS OF ARCHIVAL HARPS VELOCITIES

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ABSTRACT

Public radial-velocity archives now span nearly two decades, enough to test whether a low-amplitude signal keeps its phase over hundreds of stellar rotations. We report and test a 4.27-d signal in the HARPS velocities of the K3–K4 dwarf HD 297396 (TOI-6263). We analyse 104 HARPS-RVBank epochs over 17.9 yr with a Gaussian-process noise model, compute false-alarm and false-inclusion probabilities over 1.05–1000 d, split the data by programme, follow the signal through time, forward-simulate rotation, and test 611 other HARPS stars. The signal has $P = 4.26837 \pm 0.00027$ d and $K = 5.5 \pm 0.8$ m s^{−1}, giving $M \sin i = 11.8 \pm 1.9 M_{\oplus}$. Its false-alarm probability is 1.4×10^{-3} with all epochs (2.0 expected false alarms in the archive) and 7×10^{-6} without one discrepant night; its false-inclusion probability is of order 10^{-4} with a 201-d residual signal modelled and 0.17 without. Epochs outside the main programme recover the signal at the predicted period ($p = 1.6 \times 10^{-3}$) and phase. It holds phase to 15° over 1531 orbits, and none of 150 000 simulated rotational signals at amplitudes the data allow reproduces it, including rotation periods of 30–50 d, where the activity level and a line-width quasi-period (36–42 d) place the star. Three later HARPS spectra are consistent with the ephemeris but cannot test it. HD 297396 b is a Neptune-mass planet candidate in the period range of the Neptunian ridge. Its all-epoch significance depends on one discrepant night, and all velocities come from one instrument; one season with a second spectrograph, at a phase our ephemeris predicts to 0.1 of an orbit, will decide it.

Subject headings: planets and satellites: detection – techniques: radial velocities – stars: individual: HD 297396 – stars: late-type – methods: statistical

1. INTRODUCTION

Planets between the mass of the Earth and that of Neptune are the most common outcome of planet formation around solar-type and lower-mass stars (Mayor et al. 2011; Fulton et al. 2017). Their occurrence depends on stellar mass and metallicity (Petigura et al. 2018; Bashi et al. 2020), and K dwarfs are a good place to measure it: they are quieter than M dwarfs, and a given planet induces a larger reflex motion on them than on a solar analogue (Lillo-Box et al. 2022). They are nonetheless under-represented among radial-velocity (RV) planets with measured masses. At short periods the population has structure of its own: Neptunes avoid periods below about 3 d, the Neptunian desert, and pile up at 3.2–5.7 d, the Neptunian ridge (Castro-González et al. 2024), which high-eccentricity tidal migration may produce (Castro-González et al. 2026).

Archival data can add planets to this population. The HARPS-RVBank re-reduction (Trifonov et al. 2020; Perdelwitz et al. 2024) provides velocities for every public HARPS spectrum, and it has already yielded planets (Sreenivas et al. 2022). Archival RV signals also have a mixed record. Those later withdrawn failed in one of three ways: the sampling could manufacture them (Dumusque et al. 2012; Rajpaul et al. 2016); they lay at or near a harmonic or alias of the stellar rotation period (Anglada-Escudé et al. 2014; Robertson et al. 2015; Díaz et al. 2018; Burrows et al. 2024); or they lost phase when new data arrived (Ribas et al. 2018; Lubin et al. 2021).

A long archive can test the third directly. A Keplerian holds phase indefinitely, whereas spot patterns evolve over a few rotations.

HD 297396 is a K3–K4 dwarf at 46 pc in the HARPS volume-limited planet-search sample (Mayor et al. 2011; Frensch et al. 2023), observed from 2004 to 2021. We report a 4.27-d signal in its velocities and test it against each of the three failure modes, with phase coherence over 1531 orbits as the central test. Section 2 describes the data, Sect. 3 the star, Sect. 4 the detection and its parameters, and Sect. 5 the tests of non-planetary origins. Sections 6 and 7 cover analysis choices and other signals, and Sect. 8 places the candidate in context and gives the observations that will confirm it.

2. DATA

2.1. HARPS spectra

HD 297396 (HIP 45533, TIC 293547742, TOI-6263, Gaia DR3 5325767388444756480) was observed 108 times with HARPS (Mayor et al. 2003) on the ESO 3.6-m telescope between 2004 January and 2021 December, a baseline of 6536 d, under the 18 ESO programmes listed in the acknowledgements. Of the 108 spectra, 48 were taken under programme 072.C-0488 and 36 under its continuations, both part of the volume-limited planet search; ten come from 183.C-0972 and fourteen from four other programmes.

We use the HARPS-RVBank table of Perdelwitz et al. (2024), CDS catalogue J/A+A/683/A125, in the corrected

version of its table of velocities released on 2026 April 20¹; for this star its values are identical to those of the public RVBank Ver. 02². For every spectrum it provides the DRS and SERVAL (Zechmeister et al. 2018) velocities, each corrected for the nightly zero point, and nine line-shape and chromospheric indicators. The median formal uncertainty per nightly epoch is 1.36 m s^{-1} for SERVAL and 1.46 m s^{-1} for the DRS. We adopt SERVAL, whose template matching suits a K dwarf better than a fixed mask, and use the DRS as a cross-check of the same spectra.

The 2015 June fibre upgrade introduced an offset of about 10 m s^{-1} (Lo Curto et al. 2015). We assign four instrument labels, each with its own offset and jitter (Table 1): the two dominant programmes, the other pre-upgrade programmes, and all post-upgrade data.

2.2. How the signal was found

The signal was found in September 2026 in a periodogram survey of HARPS-RVBank. All 1328 stars with at least 20 velocities on at least 15 nights over more than 100 d were searched with a generalised Lomb–Scargle periodogram that fitted the pre- and post-2015 offsets and a linear trend, taking three peaks per star by prewhitening between 1 d and the baseline. Every peak was cross-matched with planet catalogues, including aliases and harmonics. In the triage recorded that day, 227 peaks with an analytic false-alarm probability below 10^{-4} matched no catalogued planet, 109 on 98 stars passed cuts on period, minimum mass, amplitude and number of nights, and 48 passed an activity-indicator screen. HD 297396 was chosen from those 48 in a manual review of rotation-like periods, consistency across the fibre upgrade and sampling, which followed no written rule. The search outputs are released with this paper. Every analysis choice below was made after the signal was known. We therefore quote significances for this star and add an archive-wide equivalent under a conservative assumption: that all 1328 searched stars were searched at every period between 1.0 and 1000 d with the periodogram of Sect. 4.2. The rejection criteria and decision rules used below are released with the paper; the deposit postdates the analysis.

2.3. The data set

Five instrumental and observational criteria (Table 1), none depending on model residuals, reject two spectra: the only exposure reduced with the G2 instead of the K5 mask, whose DRS line width and contrast are the extremes of the sample, and one at S/N 17.7. The remaining 106 spectra, binned by night and label, give 104 epochs.

One retained epoch is discrepant. BJD 2454922.53 (2009 April 1) lies $+52 \text{ m s}^{-1}$ from the mean in SERVAL and $+51 \text{ m s}^{-1}$ in the DRS, 9.5 times the 5.5 m s^{-1} rms of the other 103 epochs, while the spectrum taken 22 h earlier lies at -0.6 m s^{-1} . The two extractions agree on it to 1.6 m s^{-1} , which places the shift upstream of either pipeline, and all seven line-shape and chromospheric indicators lie within 1.2 robust standard deviations of their medians. It is a shift of the whole spectrum with no stellar counterpart, taken at $1''.97$ seeing against a median of $0''.90$. It passes all five criteria and is kept. Every result is given with it (104 epochs) and without it (103 epochs).

¹ <https://cdsarc.cds.unistra.fr/viz-bin/cat/J/A+A/683/A125>

² https://github.com/3fon3fonov/HARPS_RVBank

TABLE 1
INSTRUMENT LABELS AND REJECTION CRITERIA.

Label	Programmes	Era	Epochs
pre_072	072.C-0488	pre-2015	47
pre_183	183.C-0972	pre-2015	10
pre_oth	10 others	pre-2015	33
post	6 others	post-2015	14
Criterion			Rejected
C1: RVBank flag $\neq 0$			0
C2: DRS–SERVAL split $> 5\sigma$, per era			2
C3: $ \text{drift} > 3 \text{ m s}^{-1}$			0
C4: S/N < 20			1
C5: Moon $< 30^\circ$ and illumination $> 50\%$			0

One spectrum fails both C2 and C4, so two of the 108 are rejected. C2 compares the extractions within each fibre era, so that the upgrade offset alone does not trigger it.

Three HARPS spectra taken after the RVBank release (2022 March 18, 2024 March 20 and 2024 November 15) were reserved as an out-of-sample test, with its decision rule fixed before their velocities were read (Sect. 5.2).

2.4. Ancillary data

TESS (Ricker et al. 2015) observed the star in eight sectors between 2019 and 2025 (Appendix F). It carries one TESS Object of Interest, TOI-6263.01, a $1.12 \pm 0.06 R_\oplus$ candidate (from its 193-ppm depth and the stellar radius of Table 2; the TOI catalogue gives $0.93 R_\oplus$ for a smaller star) at $P = 3.0178746 \text{ d}$ detected in full-frame-image photometry (Guerrero et al. 2021; Huang et al. 2020a) at S/N 9 (Sect. 7.2). We also use Gaia DR3 astrometry (Gaia Collaboration et al. 2023), the Gaia–Hipparcos proper-motion anomaly (Kervella et al. 2022), Gemini-South/Zorro speckle imaging from 2025 January 8 (Scott et al. 2021), and ASAS-SN photometry (Shappee et al. 2014; Kochanek et al. 2017; Winecki & Kochanek 2024).

3. THE STAR

3.1. Parameters

Gaia DR3 gives a parallax of $21.6610 \pm 0.0129 \text{ mas}$ after the Lindegren et al. (2021) zero-point correction, or $d = 46.166 \pm 0.027 \text{ pc}$, with RUWE 1.049. A co-addition of the 20 highest-S/N spectra (S/N 206 at $5500 \text{ \AA}$) shows single, symmetric lines, a single-peaked cross-correlation function (FWHM 6.237 km s^{-1}) and single Ca II H and K emission cores.

For the atmospheric parameters we take the mean and spread of TIC 8.2 (4717 K; Stassun et al. 2019), Gaia GSP-Spec (4626 K) and Gaia GSP-Phot (4564 K): $T_{\text{eff}} = 4636 \pm 77 \text{ K}$ and $[\text{Fe}/\text{H}] = +0.11 \pm 0.12$. With $L_\star = 0.2275 \pm 0.006 L_\odot$ this gives $R_\star = 0.739 \pm 0.026 R_\odot$. Combining the TIC 8.2 mass ($0.760 \pm 0.087 M_\odot$) with that of Kervella et al. (2022) ($0.780 \pm 0.040 M_\odot$), with a $0.030 M_\odot$ floor for the difference between the scales, gives $M_\star = 0.776 \pm 0.047 M_\odot$ (Table 2).

3.2. Rotation

The star is chromospherically variable: against the RVBank errors, $\chi^2_\nu = 3.9$ for $\log R'_{\text{HK}}$, 7.0 for H α and 18.4 for the differential line width (ΔLW). On the Mount Wilson scale its median activity level is $\log R'_{\text{HK}} = -4.79$ ($S_{\text{MW}} = 0.49$; Gomes da Silva et al. 2021). With $B - V = 1.12$, the Noyes et al. (1984) relations give a convective turnover time of 23.9 d, a Rossby number of 1.63 and $P_{\text{rot}} = 39 \text{ d}$ (31–49 d for 0.1 dex of scatter), and the calibration of Mamajek & Hillenbrand

TABLE 2
STELLAR PROPERTIES OF HD 297396.

Parameter	Value	Source
α, δ (J2016)	09:16:47.93, -49:18:03.0	Gaia DR3
ϖ (mas)	21.6610 ± 0.0129	Gaia DR3 + ZP
d (pc)	46.166 ± 0.027	this work
RUWE	1.049	Gaia DR3
$V / G / K_s$	10.238 / 9.813 / 7.545	Hip., Gaia, 2MASS
SpT	K3–K4 V	from T_{eff}
T_{eff} (K)	4636 ± 77	this work
[Fe/H]	$+0.11 \pm 0.12$	this work
$\log g$ (cgs)	4.590 ± 0.040	this work
L_* ($L_\odot$)	0.2275 ± 0.006	TIC / FLAME
R_* ($R_\odot$)	0.739 ± 0.026	this work
M_* ($M_\odot$)	0.776 ± 0.047	this work
$B - V$	1.120 ± 0.003	Hipparcos ^a
$\log R'_{\text{HK}}$	-4.79	Mt Wilson ^b
$\log R'_{\text{HK}}$	-4.64	RVBank ^c
P_{rot} (d)	31–49 (≈ 39); 36–42	Sect. 3.2
$v \sin i$ (km s ⁻¹)	$\lesssim 1$	CCF width

^a As compiled by Gomes da Silva et al. (2021). ^b Median of the HARPS S -index series of Gomes da Silva et al. (2021), converted with the $B - V$ -based photospheric correction for which the rotation calibrations were derived. ^c PHOENIX-based photospheric correction (Perdelwitz et al. 2024), not the scale of the rotation calibrations. T_{eff} and [Fe/H] are the mean and dispersion of TIC 8.2, Gaia GSP-Spec and Gaia GSP-Phot. P_{rot} : range from the activity level; quasi-period in ΔLW and $H\alpha$. The $v \sin i$ bound allows for the instrumental and macroturbulent floor of a K dwarf.

(2008) gives 38 d. At $B - V = 1.12$ the star lies near the red edge of the samples that calibrate both relations, so the 0.1-dex range understates the uncertainty; the weighted-mean activity level (-4.81) gives 40 d (32–51 d). HARPS-RVBank lists $\log R'_{\text{HK}} = -4.64$ for the star, 0.15 dex higher, but its values use a PHOENIX-based photospheric correction, which makes cool stars appear more active than on the $B - V$ -based scale (Perdelwitz et al. 2024, their Fig. 8); used as input to these relations, they would give 28.6 d. We adopt 30–50 d. The velocity tests that depend on it were also run over 25–40, 22–40 and 17–45 d with the same results; the photometric bound of Sect. 5.8 covers 25–40 d only. Per-season periodograms stacked over seven seasons reach no indicator’s 1% level anywhere in 10–60 d; injection at 25–30 d shows that a 0.03-dex rotational modulation with its phase redrawn each season would be recovered only half the time, so the archive’s sampling, not the star, limits this search. The multidimensional Gaussian-process analysis of Yu et al. (2024), which included HD 297396 among 268 HARPS stars, did not return a rotation period for it that passed their vetting; their data end in 2015, and periods were recovered for 49 of the 268 stars, so this is weak evidence either way.

A quasi-periodic Gaussian process with the period free over 10–100 d finds a period in ΔLW (103 epochs) at 41.4 d, preferred by 14.0 in $\ln L$ over an aperiodic kernel; 40 aperiodic simulations at the same epochs never gain more than 4.6. Fitted jointly with the velocities it gives 41.5 ± 0.5 d. Programme 072.C-0488 alone prefers 41.2 d, the other programmes 36.1 d, and $H\alpha$ 38.9 d. The feature is weaker in the sparser second half of the baseline, and the two programme groups differ by 5 d. The quasi-period agrees with the activity estimate, and we take 36–42 d as a candidate rotation period, not a measurement; photometry will measure it (Sect. 8.4).

Two consequences follow. The period of HD 297396 b is 0.09–0.14 of the rotation period, so its eighth to eleventh multiples fall in the allowed range, and its ninth multiple, 38.415 d, lies inside both estimates; we test harmonics di-

rectly in Sects. 5.4 and 5.7. And the star cannot rotate at 4.27 d: its equatorial velocity would be 8.8 km s^{-1} , requiring $i_* < 6.6^\circ$ (probability 0.65%) to meet $v \sin i \lesssim 1 \text{ km s}^{-1}$; the Rossby number of 0.18 would imply $\log R'_{\text{HK}} \approx -4.3$ (Mamajek & Hillenbrand 2008) against the observed -4.79 ; and TESS finds less than 63 ppm at that period (Sect. 5.8).

4. RADIAL-VELOCITY ANALYSIS

4.1. Model

The velocities are modelled as a sum of circular Keplerians plus one offset per instrument label. Correlated stellar variability is modelled by a Gaussian process with a Matérn-3/2 kernel k , fitted jointly to the velocities and to ΔLW , the SERVAL width indicator and an established activity diagnostic (Zechmeister et al. 2018; Lafarga et al. 2021). For each series s the covariance is

$$\Sigma_{ij}^{(s)} = A_s^2 k(|t_i - t_j|; \ell) + \delta_{ij} [(\beta_s e_i^{(s)})^2 + \sigma_{g(i),s}^2], \quad (1)$$

with an amplitude A_s , a scaling β_s of the formal errors e_i and a jitter σ per label $g(i)$; the length scale ℓ is shared between the two series (Rajpaul et al. 2015). Evidences are computed by nested sampling (Skilling 2006) with dynesty (Speagle 2020), with the priors of Table A1. This model, with a Gaussian likelihood on all 104 epochs, is the adopted model. It is the least favourable to the planet of the correlated-noise models we ran, because it must absorb the discrepant night by inflating the noise of a whole programme. Our rule for choosing a noise model picks a variant with a seeing-dependent variance term; we do not adopt it because its preference comes from the discrepant night alone (19.9 nats with it, 1.1 without). The full ladder of noise models is in Appendix B.

Because the Matérn length scale starts at 50 d, rotational variability is modelled by the primary alternative: a quasi-periodic Gaussian process fitted to the same two series, the standard model for spot-driven RV variability. It prefers the planet by $\Delta \ln \mathcal{Z} = +9.71$, against $+8.87$ for the adopted model, and gives $K = 5.63 \pm 0.99 \text{ m s}^{-1}$ at its maximum-likelihood covariance ($5.47 \pm 0.83 \text{ m s}^{-1}$ without the discrepant night). With its period free over 10–100 d it places its quasi-periodic term at 42 d, carried by ΔLW , and its velocity amplitude at the lower edge of its prior (below 5.1 m s^{-1} at 95%). The two models agree on the planet.

The perspective acceleration implied by the Gaia astrometry ($0.063 \text{ m s}^{-1} \text{ yr}^{-1}$, 1.1 m s^{-1} over the baseline) is absorbed by the offsets and the long-timescale process.

4.2. Detection and false-alarm probability

Figure 1 shows the maximum-likelihood periodogram, with a free offset and jitter for each label and a sinusoid at each trial frequency. Over 1.05–1000 d its highest peak is at 4.2682 d with $\Delta\chi^2 = 32.3$ (104 epochs) or 44.0 (103 epochs). The DRS velocities peak at the same period ($\Delta\chi^2 = 27.9$ and 37.5).

Gaussian noise drawn with the fitted jitters exceeds the observed peak anywhere in the band in 14 of 10 000 draws for the 104-epoch set, a global false-alarm probability (FAP) of 1.4×10^{-3} , and in none for the 103-epoch set, where the analytic bound of Baluev (2008) gives 6.7×10^{-6} (2.2×10^{-3} for 104 epochs). Permuting the actual velocities within each label gives no exceedance in 10 000 draws for either set. Weighting the periodogram by the Matérn or quasi-periodic covariance leaves P_0 the highest peak, with FAPs of 1.2×10^{-3} and 8×10^{-4} for all epochs (Appendix B).

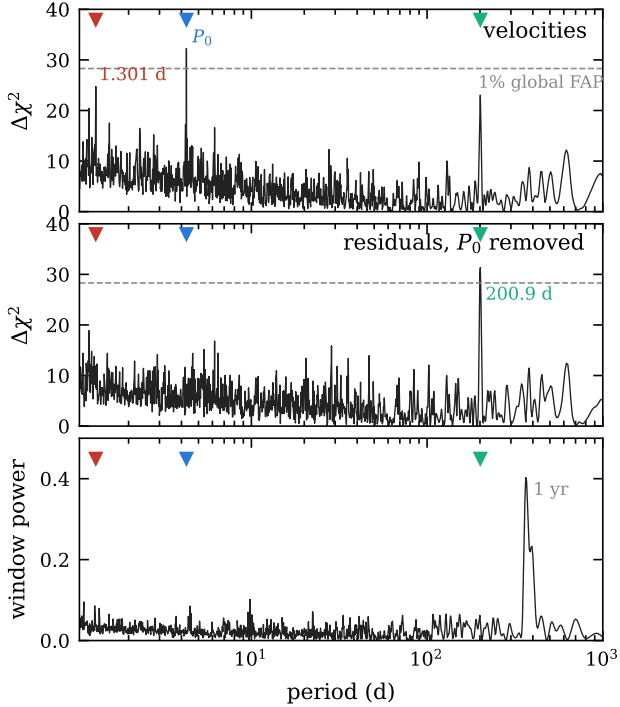


FIG. 1.— Periodograms of the 104 SERVAL epochs. *Top*: maximum-likelihood $\Delta\chi^2$ with four per-label offsets and jitters. *Middle*: the same after removing a sinusoid at P_0 . *Bottom*: the spectral window. The markers show $P_0 = 4.268$ d (blue), its sidereal-day alias at 1.301 d (red) and the residual signal at 200.9 d (green). The dashed line is the 1% global FAP from 10 000 bootstrap draws.

Multiplied by the 1328 searched stars of Sect. 2.2, and by 1.05 for the wider band of the survey (1.0 rather than 1.05 d), the FAPs give 2.0 expected false alarms of this height in the archive with all epochs and 9×10^{-3} without the discrepant night. The difference is one night: it raises the fitted jitter of programme 072.C-0488 from 3.1 to 7.7 m s⁻¹ and down-weights all 47 of its epochs (Appendix D). The first test of Sect. 5 (Fig. 2, Sect. 5.1) shows that two disjoint groups of programmes find the same signal.

4.3. Model comparison and false-inclusion probability

Under the adopted model the planet is preferred over no planet by $\Delta \ln \mathcal{Z} = +8.87 \pm 0.34$ with the narrow period prior of Table A1. Widening the prior to the full search band, a Jeffreys prior over 1.05–1000 d, costs 5.34 nats and leaves +3.53, odds of 34:1. Models that treat the discrepant night as an outlier rather than as noise give +11.7 (Student-*t*), +25.6 (one-outlier mixture) and +27.4 (night deleted) (Table B1).

The false-inclusion probability (FIP; Hara et al. 2022b) asks the search question directly: the posterior probability that no planet lies in the band 4.20–4.34 d, marginalised over the number of planets, their periods over 1.05–1000 d and all noise parameters. We computed it in the white-noise frame, where offsets and amplitudes integrate analytically on a fine frequency grid (Appendix A). That frame omits the slow correlated variability that the Gaussian process models (preferred by 24.7 nats on the planet-free model). On the narrow prior it reproduces nested sampling of the same model code to within 1 nat (+7.77) and juliet to about 2 nats.

With up to two planets allowed, the FIP of the P_0 band is $1.7\text{--}2.3 \times 10^{-4}$ with all epochs, at the resolution of the computation, and below 10^{-4} without the discrepant night. The sec-

ond planet sits at 200.9 d (Sect. 7.1), and the two-planet model is preferred by 8.5–8.6 nats. With at most one planet the FIP is 0.15–0.17 with all epochs, because that planet is then shared between P_0 (87–88% of the posterior) and 200.9 d, and 9×10^{-5} without the night. Holding the jitters at their planet-free values instead of marginalising them, which lets them absorb the signal, gives 0.45 and 0.006 for two planets. The FIP describes this star; the search that selected it is accounted for by the archive-wide FAP above.

4.4. Period and alias

The window function (Fig. 1) is dominated by the sidereal day, the year and the synodic month, and P_0 lies on none of them: the window at P_0 sits at the 22nd percentile of its own distribution. The 200.9-d residual signal of Sect. 7.1, of comparable $\Delta\chi^2$, sits at the 99.3rd percentile, next to a window maximum. These data can therefore carry a residual as strong as the candidate; what separates the two is the window and the tests of Sect. 5, which the 200.9-d signal has not been put through.

The strongest competitor of P_0 is its sidereal-day alias at 1.30131 d. With every noise parameter marginalised and Jeffreys priors of equal logarithmic width, P_0 is preferred by 5.7 nats with all epochs and 9.9 without the discrepant night (4.5 and 8.8 with priors uniform in frequency); nested sampling of the adopted model gives $+6.02 \pm 0.36$. Of 4000 simulations of a signal at 1.30131 d, none reproduces the observed preference, while 96% of simulations with the signal at P_0 pick the correct peak. P_0 also wins at every one of the 104 single-epoch deletions. No window line connects the rotation band to P_0 : its sidereal-day, synodic-month and yearly parents lie at 0.81–4.99 d.

4.5. Parameters of HD 297396 b

The adopted model gives $P = 4.26837 \pm 0.00027$ d and $K = 5.52 \pm 0.84$ m s⁻¹, and with $M_\star = 0.776 \pm 0.047 M_\odot$, $M \sin i = 11.8 \pm 1.9 M_\oplus$ at $a = 0.04734 \pm 0.00096$ au (Table 3, Figs. 3 and 4). The velocities dominate the mass error (15%, against 4% from the stellar mass). Across 32 combinations of extraction, noise model, likelihood and outlier treatment, K spans 4.91–5.72 m s⁻¹ with a standard deviation of 0.17 m s⁻¹.

We adopt a circular orbit. Without the discrepant night the eccentricity is below 0.20 (95%, Kipping 2013 prior). With it, the likelihood admits an $e \approx 0.9$, $K \approx 30$ m s⁻¹ solution that places periastron on that night and disappears without it.

TESS shows no transit. The mass–radius relation of Chen & Kipping (2017) gives $R_p \approx 3.4 R_\oplus$ and a depth of about 1800 ppm; an Earth-like composition (Zeng et al. 2019), about $2.0 R_\oplus$, gives 600 ppm. Both are three to nine times the 193 ppm of TOI-6263.01, which the same photometry detects, so the non-detection excludes $i > 85^\circ.7$ and the true mass is $14.5^{+8.5}_{-3.1} M_\oplus$.

5. TESTS OF A NON-PLANETARY ORIGIN

Table 5 lists each alternative to an orbiting body, the signature it would leave, what we measured and the verdict. This section gives the measurements.

5.1. Two disjoint groups of programmes

The 104 epochs fall into two groups that share no nights. Programme 072.C-0488 contributed 47 epochs between 2004 and 2009; the other 17 programmes contributed 57 epochs

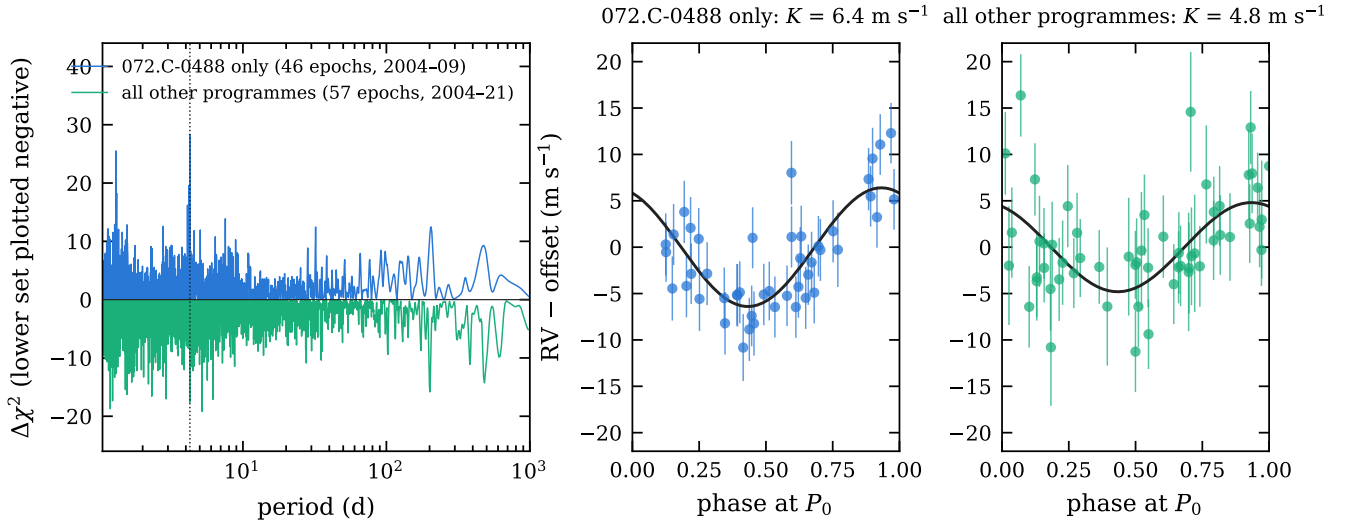


FIG. 2.— The signal in two disjoint groups of observing programmes. *Left*: periodograms of programme 072.C-0488 alone (46 epochs, 2004–2009, blue, upward) and of the 17 other programmes (57 epochs, 2004–2021, green, downward). The dotted line marks P_0 . *Centre and right*: each group phase-folded on P_0 , with its own offsets and sinusoid; the phases are $-146 \pm 6^\circ$ and $-146 \pm 10^\circ$.

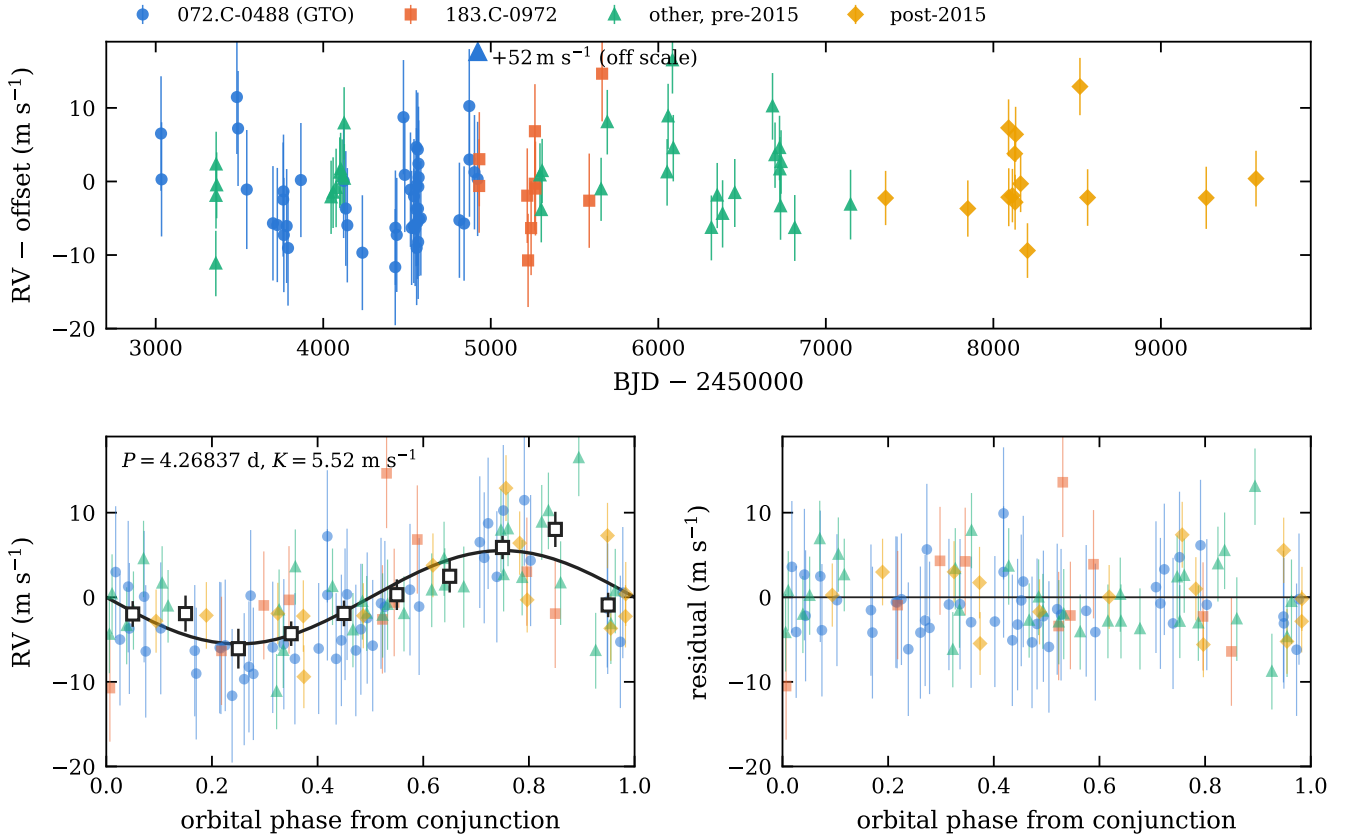


FIG. 3.— *Top*: the 104 SERVAL velocities over 17.9 yr after removal of the four fitted offsets, coloured by instrument label. The triangle marks the discrepant night, which lies above the plotted range. *Bottom left*: velocities phase-folded on P_0 , with the circular Keplerian of the adopted model and binned means (open squares). *Bottom right*: residuals on the same scale. Error bars include the fitted jitter of each label.

TABLE 3
PARAMETERS OF HD 297396 B UNDER THE ADOPTED MODEL.

Parameter	Prior	Value
P (d)	$\mathcal{J}(4.20, 4.34)$	4.26837 ± 0.00027
K (m s $^{-1}$)	$\mathcal{U}(0, 30)$	5.52 ± 0.84
T_{conj} (BJD–2450000)	$\mathcal{U}$ over one period	6298.34 ± 0.12
e	fixed	0
$M \sin i$ ($M_{\oplus}$)		11.8 ± 1.9
a (au)		0.04734 ± 0.00096
$a/R_{\star}$		13.8 ± 0.6
Insolation ($S_{\oplus}$)		102
T_{eq} (K)		883
Transit probability		7.5%
i (deg)		< 85.7 (TESS)
M_p ($M_{\oplus}$)		$14.5^{+8.5}_{-3.1}$
<i>$M \sin i$ ($M_{\oplus}$) under other treatments</i>		
Discrepant night deleted		12.0 ± 0.9
Student- t likelihood		12.1 ± 1.9
One-outlier mixture		11.9 ± 0.9

T_{eq} assumes zero albedo and full redistribution. M_p is the true mass under an isotropic inclination prior truncated by the TESS non-detection.

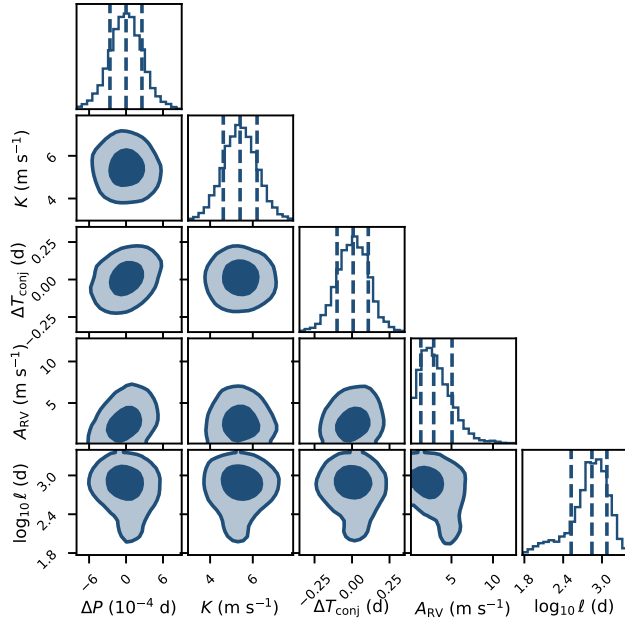


FIG. 4.— Posterior of the adopted model (104 epochs): the orbital parameters and the velocity amplitude and shared length scale of the Gaussian process. ΔP and ΔT_{conj} are offsets from the values of Table 3. The figure is from a repeat run, which gives $K = 5.40 \pm 0.79$ m s $^{-1}$, 0.15σ from the value in Table 3. Contours enclose 39% and 86% of the posterior; dashed lines mark the 16th, 50th and 84th percentiles.

between 2004 and 2021. We used each group to predict the signal and tested the prediction on the other (Fig. 2).

Programme 072.C-0488 alone (46 epochs, without the discrepant night) has its highest peak over 1.05–1000 d at 4.2692 ± 0.0010 d, with $K = 6.5 \pm 0.7$ m s $^{-1}$. At that period, within $\pm 3\sigma$, the other 57 epochs peak at 4.2682 d with $\Delta\chi^2 = 17.6$ and $K = 4.9 \pm 0.9$ m s $^{-1}$; noise produces a peak this high in that window with probability 1.6×10^{-3} (Gaussian) or 3×10^{-4} (velocities permuted within labels). In the reverse direction the 57 epochs predict 4.26820 ± 0.00037 d, and the 46 epochs give $\Delta\chi^2 = 28.4$ there, which no draw of 20 000 reaches. At P_0 the two groups give phases of $-146 \pm 6^\circ$ and $-146 \pm 10^\circ$.

Two sets of nights, taken by different programmes over different eras of the instrument, find the same period, the same phase and amplitudes within 1.4σ . The signal is not carried by one programme, one night or one era. Taken alone, the 57-epoch group has a slightly higher peak at 5.1975 d ($\Delta\chi^2 = 19.2$); the test fixes the period in advance from the other group. Both groups belong to the star that was selected, so the test establishes where the signal lives in the data rather than a selection-free significance.

5.2. Three spectra taken after the archive release

Three HARPS spectra of the star were taken after the RVBank release, on 2022 March 18, 2024 March 20 and 2024 November 15, under programmes 108.222V.001 and 106.21R4.001. They were reduced with the same pipeline (DRS 3.8), mask (K5) and fibre configuration as the last two RVBank spectra, without a simultaneous reference. They pass criteria C4 and C5 of Table 1; C1 and C2 need RVBank products, and C3 cannot be applied without a simultaneous reference. We tested them with a decision rule written into code on 2026 September 25, before their velocities were read; the code is released with this paper, but the date is not independently time-stamped (Sect. 2.2).

Because their zero point was treated as unknown, the test is a contrast. At orbital phases of 0.80, 0.52 and 0.80 from conjunction, the adopted model predicts that the 2024 March velocity lies 4.2 ± 2.9 m s $^{-1}$ below the mean of the other two. The observed contrast is -12.7 m s $^{-1}$. It is carried by the 2024 March spectrum, the one with the lowest S/N, which lies 13 m s $^{-1}$ below the other two; those agree with each other to 2.5 m s $^{-1}$. With the 3.9 m s $^{-1}$ per-epoch scatter of the fit and the pipeline errors of these spectra (3.5, 5.7 and 3.9 m s $^{-1}$), the noise on the contrast is 7.9 m s $^{-1}$. The result is consistent with the prediction ($z = -1.0$) and lies 1.6σ from zero in the predicted direction. The test had an 8% chance of excluding zero had the planet been real, against about 20% anticipated before the pipeline errors were known. With the photon-noise errors of the cross-correlation (1.3, 3.6 and 1.6 m s $^{-1}$) in place of the registered pipeline errors, a choice not made in advance, the noise is 6.0 m s $^{-1}$, $z = -1.3$ and the contrast lies 2.1σ from zero. For spectra in this fibre mode the pipeline errors are 2.5–3.4 times the SERVAL errors of the two 2021 RVBank spectra, so the registered test is the conservative one, and we report it as the result.

After the test we found that the ESO HARPS radial-velocity catalogue (HARPS_RVCAT_V1) reproduces the RVBank DRS velocities of all 108 spectra to within 0.001 m s $^{-1}$. The three new velocities therefore share the RVBank scale up to the nightly zero point, which RVBank corrects and which scatters by 0.98 m s $^{-1}$ from night to night for this star. With their offset tied to the post-upgrade DRS velocities, the nightly zero point added as noise, and a shared 2 m s $^{-1}$ allowance for any later instrumental change, the three spectra give odds of 0.75:1 between the planet and no planet. They slightly favour no planet because the 2022 and 2024 November velocities lie closer to the archive mean than the orbit predicts. This comparison was not planned before the data were read. The three spectra are consistent with the candidate but cannot test it.

5.3. Phase coherence

Spot patterns evolve over one to a few rotations, and even long-lived active longitudes drift in phase with differential rotation; a Keplerian holds phase indefinitely. With 17.9 yr of

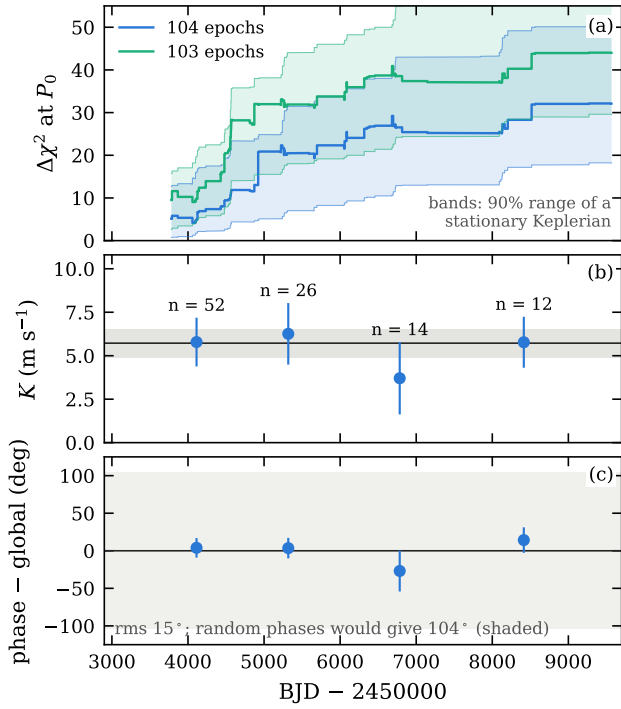


Fig. 5.— Coherence over 17.9 yr. (a) $\Delta\chi^2$ at P_0 as epochs accumulate in time order, for 104 (blue) and 103 (green) epochs; each band is the 90% range of a stationary Keplerian of the fitted amplitude and phase with the actual residuals as noise. (b) Semi-amplitude of a sinusoid at P_0 in four blocks of equal length, with the global value as a band and the number of epochs per block. (c) Block phase relative to the global phase; the shaded band is the scatter that random phases would give.

data, coherence is the sharpest discriminant available, and we measure it three ways.

First, we follow the signal as epochs accumulate in time order (Mortier & Collier Cameron 2017; Sreenivas et al. 2022). Figure 5a shows $\Delta\chi^2$ at P_0 after each new epoch against the 90% range of 2000 simulations of a stationary Keplerian of the fitted amplitude and phase, added to the fit residuals permuted within each label. The observed curve lies inside that range at every step, for both data sets; its mean squared deviation from the simulated mean is exceeded by 88% and 83% of the simulations.

Second, we fitted a sinusoid at P_0 in four blocks of equal length (52, 26, 14 and 12 epochs; Fig. 5b,c). The block phases scatter by 15° rms about the global phase, against 104° for random phases. A constant phase has χ^2 probability 0.61 and a constant amplitude 0.79. The block amplitudes do not follow the block-mean activity level, a weak test because the block means span only 0.16 dex in $\log R'_{\text{HK}}$.

Third, we replaced the constant amplitude with a Gaussian envelope $K \exp[-(t - t_a)^2/2\tau^2]$, a form of the apodised-sinusoid test of Hara et al. (2022a), and profiled the likelihood over t_a . The coherence time is $\tau > 2300$ d (95%, one-sided); more than 46 rotations even at 50 d, the long edge of the adopted range, and 540 orbits. Simulated spotted stars produce signals that decay over a few rotations (Boisse et al. 2011).

5.4. What rotation can produce

We asked the reverse question directly: how often does rotational activity, at the largest amplitude these data allow, produce what we observe? Each realisation is a strictly coherent

sinusoid at a period drawn from 30–50 d, with its first two harmonics at one half and one third of the fundamental amplitude, held in phase for the full 17.9 yr, a null stronger in its coherence than any spot model. The fundamental amplitude is 4.5 m s^{-1} , the 90%-recovery bound of the harmonic scan (Sect. 5.7). The noise is the actual residuals permuted within labels, and each realisation passes through the same periodogram as the data.

None of 20 000 realisations reaches the observed $\Delta\chi^2 = 32.3$ at P_0 (Fig. 6); the 99th percentile is 9.0 and the maximum 20.1. A fundamental of 12 m s^{-1} reaches it in 0.2% of realisations and 15 m s^{-1} in 2.2%, amplitudes two to three times the candidate’s that would dominate the velocities, the indicators and the photometry. The same procedure applied to a Keplerian at the fitted amplitude succeeds in 72% of trials, so the test has power. Rotation periods of 25–40, 22–40 or 17–45 d give the same result.

Evolving spots do no better. Realisations of a quasi-periodic Gaussian process, $k(\tau) = A^2 \exp[-\tau^2/2\lambda^2 - \sin^2(\pi\tau/P_{\text{rot}})/2w^2]$, with rotation period 17–45 or 30–50 d, evolution time λ of one to ten rotations and harmonic complexity w spanning that fitted to ΔLW , never reach the observed value: none of 90 000 at 3.2 or 6 m s^{-1} rms. At 10 m s^{-1} rms, almost twice the total velocity scatter of the star, 0.3–0.4% do.

A rotation period at $9P_0 = 38.415$ d, inside both rotation estimates, needs no leakage: the ninth harmonic of a strictly periodic signal falls on P_0 . Two properties of the data argue against it. The phase would have to hold to 15° over 170 rotations, which requires a rotation period constant to 3 parts in 10^5 (Sect. 5.7); and the ninth harmonic would have to carry $K \approx 5.5 \text{ m s}^{-1}$ while the fundamental ($0.5 \pm 0.8 \text{ m s}^{-1}$) and every other harmonic of 38.415 d up to the eleventh carry less than 3.0 m s^{-1} at 95% confidence in the residuals of the planet fit. Spot models do not distribute power this way: in the kernel above even $w = 0.15$ puts 0.7% of the variance into the ninth harmonic, so $K = 5.5 \text{ m s}^{-1}$ there would need a rotational signal of about 50 m s^{-1} rms. Realisations with P_{rot} within 1% of $9P_0$ and $w = 0.15$ –0.4 behave like the rest of the band (none of 20 000 at 3.2 or 6 m s^{-1} rms, 0.4% at 10 m s^{-1}), which confirms this budget rather than testing a coherent ninth harmonic. This is the forward simulation that retired the 3.2-d signal of α Cen B (Dumusque et al. 2012; Rajpaul et al. 2016), applied to this candidate.

5.5. Activity indicators

None of the nine HARPS-RVBank indicators has power at P_0 (Fig. 7, Table 4): the periodogram value there is 6–28% of each indicator’s own 1% false-alarm level. The only indicator peak above its level is in ΔLW near 320 d, where the Gaussian process places the correlated noise. The largest correlation between an indicator and the velocity residuals is $r = 0.22$ ($H\alpha$, $p = 0.024$), which does not survive a Bonferroni threshold of 0.0056 for nine tests.

5.6. A cross-star control

An instrumental or pipeline signal at 4.27 d would appear in other stars. We computed the same statistic at exactly 4.26838 d for every HARPS-RVBank star with at least 30 nightly epochs over more than 1000 d, with separate offsets and jitters before and after the fibre upgrade, and kept the 611 stars whose median $\Delta\chi^2$ at 40 nearby control frequencies is below 3. Their median at P_0 is 1.21, against 1.26 at the control

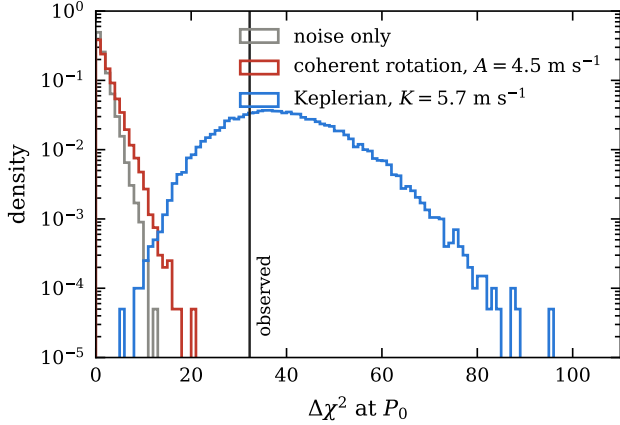


FIG. 6.— $\Delta\chi^2$ at P_0 for 20 000 realisations each of noise alone, of strictly coherent rotational modulation at the largest amplitude the data permit (rotation period drawn from 30–50 d), and of a Keplerian at the fitted amplitude, sampled at the real epochs with the real residuals as noise. The observed value is marked.

TABLE 4
THE NINE ACTIVITY INDICATORS AT P_0 .

Indicator	n	power/1%	r	p
ΔLW	104	0.06	+0.09	0.36
$\text{H}\alpha$	104	0.12	+0.22	0.024
Na I D1	104	0.09	+0.01	0.89
Na I D2	104	0.13	+0.02	0.87
CCF FWHM	104	0.14	+0.08	0.40
CCF contrast	104	0.06	−0.05	0.62
BIS	104	0.28	+0.03	0.75
CRX	104	0.16	−0.07	0.47
$\log R'_{\text{HK}}$	98	0.19	−0.02	0.82

Column 3 is the periodogram value at P_0 over the indicator’s own 1% false-alarm level. Columns 4 and 5 are the Pearson correlation with the velocity residuals of the adopted fit and its two-sided p value.

frequencies and 1.39 expected for pure noise. The sample is not matched to HD 297396 in spectral type or mask, so it tests effects common to the instrument and pipeline. None exceeds 13.8, where a χ^2_2 distribution expects 0.6 stars; HD 297396 sits at 32.3–44.0. Among the 54 stars that share at least ten nights with it, the largest value is 8.1. The signal is also present in the velocities before the nightly zero-point correction ($\Delta\chi^2 = 31.9$ and 44.6), and the correction itself carries only $0.18 \pm 0.12 \text{ m s}^{-1}$ at P_0 . No systematic common to the instrument, the pipeline or the zero-point correction produces power at this period; an effect specific to this star’s spectra is left for a second instrument.

5.7. Rotational harmonics

Spot-induced velocities concentrate at P_{rot} and its first two harmonics (Boisse et al. 2011; Vanderburg et al. 2016); higher harmonics appear mainly for low-latitude spots and faster rotators (Barnes et al. 2024), and active regions can produce peaks with no harmonic relation to P_{rot} (Nava et al. 2020), so the harmonic index alone does not decide the question. The quasi-periods measured in Sect. 3.2 (38.9, 41.4 and 36.1 d) do not place a harmonic at P_0 : the nearest lie 18, 47 and 83 frequency-resolution elements away. Only $P_{\text{rot}} = 38.41 \pm 0.03 \text{ d}$ would put one within a single element, and that value lies inside both the activity range and the spread of the quasi-periods, so the period alone does not exclude it

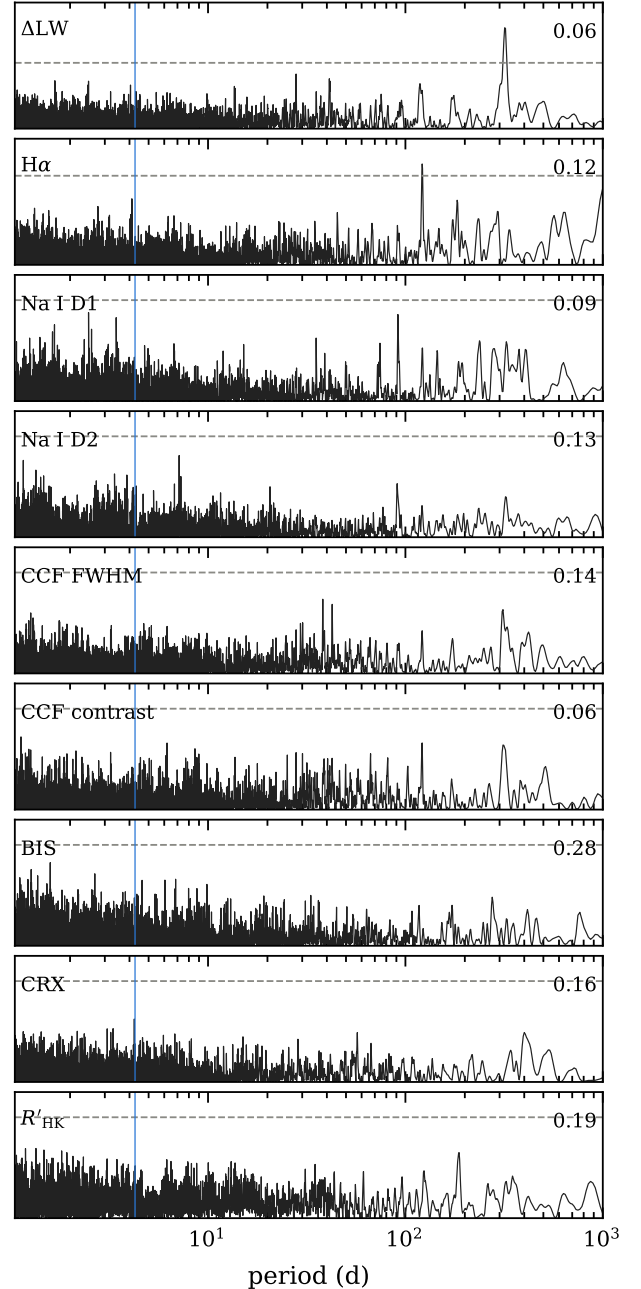


FIG. 7.— Periodograms of the nine activity indicators on the velocity epochs, each with four per-label offsets. The dashed line is each indicator’s 1% global false-alarm level from within-label shuffles, and the number at right is the ratio of the periodogram value at P_0 (vertical line) to that level.

(Sect. 5.4). Phase stability is the quantitative test. Holding 15° of phase over 1531 cycles requires the underlying period to stay constant to 3 parts in 10^5 for 17.9 yr, about 1.5 minutes for a 39-d rotation period, whereas solar surface rotation differs by 5% between the equator and 30° latitude (Snodgrass & Ulrich 1990).

We also searched the residuals of the Keplerian fit directly. The largest $\Delta\chi^2$ is 13.9 over 30–50 d, 13.7 over 15–25 d and 11.0 over 10–16.7 d, the fundamental and first two harmonics of the adopted rotation range, against band-restricted 1% levels of 20.2, 21.3 and 22.0. No band is populated, and a sinusoid of $K = 4.5 \text{ m s}^{-1}$ anywhere in 30–50 d would have

been recovered nine times in ten. The bands of 25–40, 22–40 and 17–45 d give the same result. The same bands do exceed their levels in the raw velocities, but synthetic data containing only the Keplerian reproduce that excess: it is window leakage from the planet, which is why we test the residuals.

5.8. Photometry

Eight TESS sectors exclude a photospheric modulation at P_0 . In every sector the amplitude at P_0 lies below its 99th-percentile null, the sector phases are consistent with uniform (Rayleigh $p = 0.46$), and the stitched light curve bounds the amplitude at 63 ppm (3σ , corrected for the detrending filter). Spots below that level produce RV amplitudes of order 0.06 m s^{-1} , 90 times below the observed semi-amplitude.

The ASAS-SN aperture photometry shows an $11.9 \pm 3.1 \text{ mmag}$ modulation at P_0 , which TESS excludes by a factor of 60–170 and which the saturation-corrected reduction of Winecki & Kochanek (2024) does not reproduce ($< 5.5 \text{ mmag}$, 3σ); it is an artefact of a saturated star (Appendix E). That reduction bounds any modulation in 25–40 d below 8 mmag; periods of 40–50 d were not searched.

5.9. Companions and detection limits

Four probes exclude a stellar companion over more than three decades in separation: Gaia astrometry (RUWE 1.049); the Gaia–Hipparcos proper-motion anomaly (S/N 1.09, limiting companions to $\approx 1.2 M_{\text{Jup}}$ at 3–10 au; Kervella et al. 2022); Zorro speckle imaging at 832 nm ($\Delta m = 7.0$ at $0''.5$, 23 au, and 8.3 at the outer edge of the sensitivity curve, $1''.18$ or 54 au; 5.6 at $0''.5$ at 562 nm); and the absence of an RV acceleration ($+0.08 \pm 0.18 \text{ m s}^{-1} \text{ yr}^{-1}$). The single-lined spectrum and single CCF peak exclude a double-lined binary. Together these exclude stellar companions from 0.02 au to the edge of the speckle sensitivity curve at 54 au. At 1.8–3 au, between the astrometric probes, a stellar or brown-dwarf companion would produce hundreds of m s^{-1} over the baseline.

Injection into the residuals of the white-noise fit, at 36 periods from 2 to 1000 d and 60 phases each, recovers $K = 5\text{--}6 \text{ m s}^{-1}$ signals 95% of the time over 2–200 d, rising to $6\text{--}9 \text{ m s}^{-1}$ beyond 200 d. At P_0 , 95% recovery needs $K \approx 4.6 \text{ m s}^{-1}$ in this most conservative noise frame.

6. DEPENDENCE ON ANALYSIS CHOICES

Two choices change how strongly the signal is detected, the treatment of the discrepant night and the extraction; neither moves the planet. Every treatment of the night favours the planet under the narrow prior, with $\Delta \ln \mathcal{Z}$ from +6.5 to +27.4; over the full band the weakest cells, the adopted model with DRS velocities and white noise with all epochs, are indecisive. Across all treatments, K changes by at most 0.15σ (5.52 to 5.64 m s^{-1}). Deleting each of the 104 epochs in turn changes K by 0.10 m s^{-1} rms, and the discrepant night itself moves it by 0.03 m s^{-1} . The two extractions agree on K within 0.2σ (5.52 ± 0.84 and $5.72 \pm 1.03 \text{ m s}^{-1}$); the DRS needs larger jitters and gives a lower evidence under the adopted model (Table B1). The night sets the precision and the size of the evidence, not the existence or the mass of the planet (Appendices B and D).

7. OTHER SIGNALS IN THE DATA

7.1. A residual signal at 200.9 d

After the Keplerian at P_0 is removed, the highest peak in every data set is at 200.9 d (Fig. 1), with $\Delta\chi^2 = 31.6$, a global

FAP of 0.2% and $K_2 = 4.9 \pm 0.6 \text{ m s}^{-1}$ in the white-noise frame. It survives trend removal to cubic order, no activity indicator has power there, and fitting it changes the semi-amplitude of b by 0.25σ . Our promotion rule requires a second signal to win by more than 6 nats and to lie off the peaks of the spectral window. Under the adopted model it wins by $+5.98 \pm 0.40$ nats (Jeffreys prior on 150–260 d) and by +3.5 over the full band, and it lies 0.25 resolution elements from a window maximum at 199.355 d, at the 99.3rd percentile of the window. The barycentric velocity, annual by construction, shows $\Delta\chi^2 = 43$ at this period through leakage alone. We label it an unresolved residual signal. Twenty epochs at the edges of the observing season would break the window peak and resolve it (Sect. 8.4).

7.2. TOI-6263.01

The TESS candidate at 3.0178746 ± 0.0000265 d has a depth of 193 ± 17 ppm at S/N 9 in the QLP photometry, and the TOI catalogue notes a low S/N and a slight dependence of depth on aperture; its disposition is planet candidate. Ground-based photometry from the TESS Follow-up Observing Program (LCO-CTIO 1 m, 2023 April 25) cleared 137 of 139 neighbours within $2''.5$ as sources of the signal but could not detect a 0.19-ppt event on target. Its semi-amplitude predicted from its radius with the relation of Chen & Kipping (2017) is 0.8 m s^{-1} . A Keplerian at its ephemeris gives $K = 1.3 \text{ m s}^{-1}$ with a 95% upper limit of 2.7 m s^{-1} ($M \sin i < 5.2 M_{\oplus}$), and including it does not change K_b . If both are planets, their separation is 8.9 ± 0.5 mutual Hill radii, above the Gladman (1993) criterion in every one of 2×10^5 Monte Carlo draws, and the period ratio is 1.414. The designation HD 297396 b follows the order of announcement and implies nothing about the letter of TOI-6263.01.

8. DISCUSSION

8.1. The planet in context

The Neptunian ridge. Neptunes are rare at periods below about 3 d and over-abundant at 3.2–5.7 d, the Neptunian ridge between the desert and the savanna (Castro-González et al. 2024). At 4.27 d HD 297396 b lies in the middle of the ridge in period. The ridge was mapped in period–radius space for planets of 4–10 $R_{\oplus}$. HD 297396 b does not transit; the relation of Chen & Kipping (2017) places it at 3.4 and 3.8 $R_{\oplus}$ for its minimum and expected true mass, just below that range. It therefore samples the low-mass edge of the ridge population, around a K dwarf, a host type under-represented among planets with measured masses.

A test of how the ridge forms. Castro-González et al. (2026) show that high-eccentricity tidal migration, triggered by massive outer companions, reproduces the ridge. A planet migrating that way crosses the orbits of any inner planets on its way in, and for migrating giants this destroys or ejects them (Mustill et al. 2015). If TOI-6263.01, a low-S/N TESS candidate (Sect. 7.2), is confirmed, HD 297396 hosts an Earth-sized transiting planet at 3.02 d inside a non-transiting Neptune-mass planet at 4.27 d, with a period ratio of 1.41. If that result carries over to Neptunes, an intact inner companion favours a quiescent origin, disc migration or formation in place, for this planet, and the system becomes a direct test of the migration picture. The geometry adds a constraint: TOI-6263.01 transits only if its orbital inclination exceeds $84^\circ.7$, and b escapes transit only if its inclination is below $85^\circ.7$. For the mutual inclinations of 1–2° typical of compact multi-planet systems

TABLE 5

THE ALTERNATIVES TO A PLANET: THE SIGNATURE EACH WOULD LEAVE, WHAT WAS MEASURED AND THE VERDICT.

Alternative	Signature	Measured	Verdict	Sect.
Noise fluctuation	No periodic signal	Highest peak over 1.05–1000 d; FAP 1.4×10^{-3} (104 epochs; 2.0 archive-wide), 7×10^{-6} (103); FIP $\sim 10^{-4}$ (104) and $< 10^{-4}$ (103) with two planets allowed, 0.15–0.17 and 9×10^{-5} with one; $\Delta \ln \mathcal{Z} = +8.87$ adopted	disfavoured	4.2, 4.3
One programme or one night	Signal confined to a subset	Highest peak of 072.C-0488 alone; recovered at that period in the 57 epochs of the 17 other programmes, same phase, K within 1.4σ ; $p = 1.6 \times 10^{-3}$ held out	excluded	5.1
Daily alias	Power at P_0 as an image of 1.301 d	Matched-prior $\Delta \ln \mathcal{Z} = +6.02$; 0 of 4000 alias simulations reproduce the preference; wins all 104 single-epoch deletions	excluded	4.4
Aliased activity	Rotational signal mapped onto P_0	No window line connects 30–50 d to P_0 ; parents at 0.81–4.99 d	excluded	4.4
Rotation at P_0	$P_{\text{rot}} = 4.27$ d	Needs $i_* < 6^\circ$ (0.65%) and Rossby number 0.18, inconsistent with $\log R'_{\text{HK}} = -4.79$; TESS < 63 ppm at P_0	excluded	3.2
Spectroscopic activity	Power at P_0 in line shape or chromosphere	6–28% of the 1% level in nine indicators; no residual correlation survives Bonferroni. The null is calibrated in the rotation band, so this is a bound	bounded	5.5
Rotational modulation	Decay on the spot-evolution timescale	Phase rms 15° over 17.9 yr ($p = 0.61$); $\tau > 2300$ d, more than 46 rotations at 50 d; grows as a stationary Keplerian	excluded	5.3
Coherent rotation leaking to P_0	Power at P_0 from activity at 30–50 d	0 of 20 000 strictly coherent realisations at 4.5 m s^{-1} (also for 25–40 d); 2.2% at 15 m s^{-1}	excluded	5.4
Rotational harmonic	Power at P_{rot} and its harmonics	Residual bands at 0.5–0.8 of their 1% levels for 30–50 and 17–45 d; $K < 4.5 \text{ m s}^{-1}$ in 30–50 d at 90% recovery; none of 110 000 quasi-periodic realisations up to 6 m s^{-1} rms reproduces the signal; at $P_{\text{rot}} = 9P_0$ the ninth harmonic would have to exceed the fundamental and every other harmonic ($K_{95} \leq 3 \text{ m s}^{-1}$)	bounded	5.4, 5.7
Instrument or pipeline	Same period in other stars	611 stars: median $\Delta\chi^2 = 1.21$ at P_0 , none above 13.8; zero-point correction carries 0.18 m s^{-1} ; star-specific effects not tested	excluded for the instrument	5.6
Photometric modulation at P_0	Spots varying at P_0	Eight TESS sectors, < 63 ppm; ASAS-SN re-reduction < 5.5 mmag	excluded	5.8
Photometric modulation at P_{rot}	Spots varying in 30–50 d	ASAS-SN g , < 8 mmag at 90% recovery in 25–40 d; 40–50 d not searched	bounded	5.8
Bound companion or blend	Diluted or composite reflex motion	Excluded from 0.02 to 54 au by astrometry, PMa, speckle and the velocities; single-lined, single-peaked CCF	excluded	5.9

Excluded: ruled out by the measurement in column 3. *Bounded:* not ruled out; the bound is given. A measured rotation period decides the two bounded rotational rows.

(Fabrycky et al. 2014), the true mass of b is then within 1% of its minimum mass.

A well-defined target. At $V = 10.2$ the star is within easy reach of ESPRESSO and NIRPS, and the ephemeris of Table 3 fixes the phase a second spectrograph should find, so it tests a prediction rather than repeating a search (Sect. 8.4).

8.2. Comparison with published detections

Table 6 sets HD 297396 b beside four published planets: HD 22496 b (Lillo-Box et al. 2021), a short-period low-mass planet around a K dwarf detected with ESPRESSO; HD 105779 b and HD 103891 b (Sreenivas et al. 2022), found in the same HARPS-RVBank archive with the same extraction; and HD 47186 b (Bouchy et al. 2009), a 4.08-d Neptune-mass planet detected with HARPS alone around a star whose rotation period was inferred from $\log R'_{\text{HK}}$.

The two HARPS-RVBank planets share our instrument, archive and noise level (residual rms $3.5\text{--}4.0 \text{ m s}^{-1}$, against 4.2 m s^{-1} here without the discrepant night). Their signals are two and four times larger and lie beyond 100 and 47 times the rotation period, where rotation cannot reach them; they were validated with white-noise models, indicator periodograms and signal growth. HD 297396 b lies at 0.09–0.14 of the rotation period, close to HD 47186 b ($P/P_{\text{rot}} \approx 0.12$) and below

HD 181433 b (0.17; Bouchy et al. 2009), both published from HARPS with an inferred rotation period, the latter retained when more data were added (Horner et al. 2019); both were detected at $K/\sigma_K > 12$ with residuals near 1 m s^{-1} around quieter stars. In that regime period separation alone does not exclude activity, so this analysis measures coherence, forward-simulates rotation and adds a cross-star control, tests that the larger, longer-period archival signals did not need.

HD 22496 b shows what the second data set provides. Its rotation period was measured from the same velocities and line widths, and archival HARPS velocities added a second instrument to $0.18\text{--}0.25 \text{ m s}^{-1}$ ESPRESSO data. Those are the two rows of Table 6 that the HARPS archive cannot supply for HD 297396. The two data sets are complementary: ESPRESSO measured the semi-amplitude to 8%, while 104 HARPS epochs over 1531 orbits bound the coherence time at more than 2300 d.

8.3. Why archival signals fail, and where this one stands

Table 7 lists four well-documented low-mass RV signals later attributed to sampling or activity. Each failed one of three screens: a semi-amplitude small enough that the window and noise could manufacture it; a period at a low-order harmonic or alias of the rotation period; or a phase that did

TABLE 6
HD 297396 b AND PUBLISHED PLANETS FROM THE SAME ARCHIVE OR IN THE SAME REGIME.

	HD 22496 b Lillo-Box et al. (2021)	HD 105779 b Sreenivas et al. (2022)	HD 103891 b Sreenivas et al. (2022)	HD 47186 b Bouchy et al. (2009)	HD 297396 b this work
Host	K5–K7 V, 0.68 $M_{\odot}$	G2 V, 0.89 $M_{\odot}$	F8–F9, slightly evolved, 1.28 $M_{\odot}$	G5 V, 0.99 $M_{\odot}$	K3–K4 V, 0.78 $M_{\odot}$
$\log R'_{\text{HK}}$ (Mt Wilson)	...	−4.89	−5.12	−5.07	−4.79
Velocities	41 ESPRESSO + 43 HARPS	53 HARPS (archive)	90 HARPS (archive)	66 HARPS	104 HARPS epochs (archive)
Median RV error (m s ^{−1})	0.18 (ESPRESSO)	1.86	1.69	0.3–0.6 (range)	1.36
Residual rms (m s ^{−1})	0.28 (ESPRESSO)	3.53	3.96	0.91	4.2 (6.3 with the discrepant night)
Baseline	895 d (ESPRESSO); 17.4 yr with HARPS	15.1 yr	14.2 yr	4.3 yr	17.9 yr
P (d)	5.09071 ± 0.00026	2412 ± 54	1919 ± 16	4.0845 ± 0.0002	4.26837 ± 0.00027
K (m s ^{−1})	2.62 ^{+0.23} _{−0.21}	10.42 ± 0.96	21.12 ± 0.86	9.12 ± 0.18	5.52 ± 0.84
K/σ_K	11.9	10.9	24.6	51	6.6 (15 without the discrepant night)
$M \sin i$	5.57 ^{+0.73} _{−0.68} $M_{\oplus}$	0.64 ± 0.06 M_{Jup}	1.44 ^{+0.06} _{−0.05} M_{Jup}	22.8 $M_{\oplus}$	11.8 ± 1.9 $M_{\oplus}$
e	< 0.15 (95%)	< 0.16 (1 σ)	0.31 ± 0.03	0.038 ± 0.020	< 0.20 (95%, without the discrepant night)
$\Delta \ln \mathcal{Z}$, planet vs none	+14.5 (+10.5 ESPRESSO alone)	+35.1	+94.3	...	+8.9 adopted; +9.7 QP GP; +25.6 one-outlier mixture
Period prior for $\Delta \ln \mathcal{Z}$	$\mathcal{U}(5.09, 1.0)$ d	$\mathcal{U}(2200, 2700)$ d	$\mathcal{U}(1800, 2000)$ d	...	$\mathcal{J}(4.20, 4.34)$ d; full band +3.5
False-alarm probability	...	< 10 ^{−3}	< 10 ^{−3}	...	1.4 × 10 ^{−3} ; 7 × 10 ^{−6} without the night
False-inclusion probability	...	...	...	...	~ 10 ^{−4} (two planets allowed); 0.15–0.17 (one); < 10 ^{−4} without the night
P_{rot}	35.0 ± 0.6 d, measured	< 23.8 d ($v \sin i$)	< 40.3 d ($v \sin i$)	33 d, from $\log R'_{\text{HK}}$	31–49 d from $\log R'_{\text{HK}}$; 36–42 d in ΔLW , $\text{H}\alpha$
P/P_{rot}	0.15	> 100	> 47	0.12	0.09–0.14
Activity model	QP GP, FWHM proxy	white noise	white noise	none (no jitter term)	Matérn GP with ΔLW ; QP GP
Second data set	HARPS added to ESPRESSO	none	none	none	none (two programme groups, Sect. 5.1; three later HARPS spectra, Sect. 8.4)
Signal growth, coherence	...	$\Delta \ln L$ grows with N , except ≤ 3 epochs	$\Delta \ln L$ grows with N	...	growth as a stationary Keplerian; 15° over 1531 orbits; $\tau > 2300$ d
Rotation forward model	...	...	...	...	0 of 150 000
Cross-star control	...	...	...	...	611 stars, none
Transit search	TESS, 4 sectors	...	...	...	TESS, 8 sectors
Companion exclusion	Gaia PMa	no Gaia PMa	no Gaia PMa	...	Gaia, PMa, speckle, RV trend
Detection limits	injection, HZ	...	...	...	injection, 2–1000 d
Designation	planet	planet	planet	planet	candidate

Entries marked ... are not reported in that work. $\log R'_{\text{HK}}$ is the median of Gomes da Silva et al. (2021) for every star (HD 22496 is not in that catalogue). Evidences from different codes and priors are not interchangeable at the nat level. All reported evidences use a period prior centred on the signal; for HD 22496 b that prior is a 1-d-wide Gaussian, and Lillo-Box et al. (2021) also report strong evidence with $\mathcal{U}(1.1, 20)$ d. Sreenivas et al. (2022) chose their priors, including $\mathcal{U}(5, 15)$ and $\mathcal{U}(18, 25)$ m s^{−1} for K , as those that maximised $\Delta \ln \mathcal{Z}$; ours is $\mathcal{U}(0, 30)$ m s^{−1}. For HD 105779 b the DRS velocities give $\Delta \ln \mathcal{Z} \lesssim 20$ (Sreenivas et al. 2022). For HD 297396 b the full-band evidence applies the prior-width correction of Sect. 4.3, and K/σ_K is also 15 under the one-outlier mixture of Appendix C.

not hold. HD 297396 b passes all three. Its semi-amplitude is four times the median formal error but comparable to the per-epoch scatter once jitter is included, so we rely instead on a forward simulation of the window and noise at the real epochs, which does not produce it (Sect. 5.4); its period is not a low-order fraction of the rotation period; and its phase holds for 17.9 yr. The screens describe how past signals failed rather than every way a signal can fail. The alternative they do not reach is an effect specific to this star in HARPS spectra, which a second instrument addresses.

8.4. What will confirm it

No ESPRESSO, UVES or NIRPS observation of this star exists, FEROS (5–10 m s^{−1} stability; Brahm et al. 2017) cannot test a 5.5 m s^{−1} signal, and the astrometric signature is 0.05 μas . Three observations will settle the candidate, in order of cost.

One season of photometry. Forward simulation shows that 25 nights of 2-mmag photometry over two to three months recover a 5-mmag modulation anywhere in 30–50 d in 95–99% of trials (at 3 mmag per night, 40 nights over four months are

needed); the spectroscopic equivalent needs more than 120 epochs. Gaia DR4 epoch photometry is the free route. A measured rotation period decides the two bounded rotational rows of Table 5.

Twenty epochs at the edges of the observing season. These break the annual window peak that blocks the 200.9-d signal and resolve it.

One season with a second spectrograph. With the 3–4 m s^{−1} stellar jitter fitted here, 40 epochs from ESPRESSO or NIRPS detect a 5.5 m s^{−1} signal at 6–8 σ on their own ($\sigma_K \approx \sigma \sqrt{2/N}$), independently of HARPS and of the discrepant night, and dense sampling within one season lets the rotation be modelled directly. The ephemeris predicts the phase to within 0.1 of an orbit (1 σ) through 2028.

The three later HARPS spectra (Sect. 5.2) are consistent with the ephemeris but, at their precision, cannot test it.

9. SUMMARY

1. A radial-velocity signal at $P = 4.26837 \pm 0.00027$ d with $K = 5.52 \pm 0.84$ m s^{−1} is present in 17.9 yr of HARPS spectra of HD 297396, giving $M \sin i = 11.8 \pm 1.9 M_{\oplus}$.

TABLE 7
LOW-MASS RV SIGNALS LATER ATTRIBUTED TO SAMPLING OR ACTIVITY, AND THE SCREEN EACH FAILED.

Signal	P (d)	What undid it	HD 297396 b on the same screen
α Cen B b (Dumusque et al. 2012)	3.24	$K = 0.5 \text{ m s}^{-1}$; reproduced by the sampling and noise alone (Rajpaul et al. 2016)	$K = 4\times$ the median formal error, comparable to the scatter; 0 of 20 000 forward simulations (Sect. 5.4)
Kapteyn b (Anglada-Escudé et al. 2014)	48.6	One third of $P_{\text{rot}} = 143 \text{ d}$, correlated with activity (Robertson et al. 2015); no planets under a GP (Bortle et al. 2021)	Harmonic index 8–11 if at all; harmonic bands empty; $P_{\text{rot}} = 9P_0$ simulated (Sects. 5.4, 5.7)
HD 26965 b (Díaz et al. 2018)	42	At the rotation period; activity lags the RV (Burrows et al. 2024)	$P_0/P_{\text{rot}} = 0.09\text{--}0.14$; $P_{\text{rot}} = P_0$ excluded (Sect. 3.2)
Barnard b (Ribas et al. 2018)	233	Transitory in new data; yearly alias of the 145-d rotation (Lubin et al. 2021)	Phase rms 15° over 17.9 yr; $\tau > 2300 \text{ d}$ (Sect. 5.3)

It is the highest peak over 1.05–1000 d, with a FAP of 1.4×10^{-3} from all 104 epochs (2.0 expected false alarms across the archive) and 7×10^{-6} without one discrepant night. The planet is preferred by $\Delta \ln \mathcal{Z} = +8.9$ under the adopted Matérn process, +9.7 under a quasi-periodic process and +3.5 over the full search band. Its FIP is $\sim 10^{-4}$ when a 200.9-d residual signal is also modelled and 0.15–0.17 when it is not (0.45 if the jitters are held at their planet-free values).

- Two disjoint groups of programmes, the 2004–2009 guaranteed-time data and the 57 epochs of 17 other programmes, each show the signal at the same period and phase, with amplitudes within 1.4σ . None of 611 other HARPS stars shows power at this period. Three HARPS spectra taken after the archive release are consistent with the ephemeris on a test fixed before their velocities were read ($z = -1.0$), but the test had an 8% chance of excluding zero and cannot decide.
- The signal behaves as a Keplerian and not as rotation. It grows with the number of epochs as a stationary Keplerian does, holds phase to 15° over 1531 orbits with a coherence time above 2300 d, and none of 150 000 simulated rotational signals, coherent or evolving, at amplitudes the data allow, reproduces it. A rotation period of exactly nine times P_0 is not excluded by its period alone; it would need a ninth harmonic stronger than the fundamental, held in phase for 170 rotations. Nine activity indicators and TESS show nothing at P_0 .
- HD 297396 b is a Neptune-mass planet candidate around a K dwarf, in the period range of the Neptunian ridge. If the low-S/N TESS candidate TOI-6263.01 is confirmed, it has an inner companion at a period ratio of 1.41, an architecture that tests high-eccentricity migration as the origin of the ridge.
- It is a candidate for three reasons, in order of weight: its all-epoch significance depends on how one night is treated; all velocities come from one instrument; and the rotation period is estimated ($\approx 39 \text{ d}$ from the Mount Wilson activity level, with a quasi-period at 36–42 d in the line widths) rather than measured. One season with a second spectrograph addresses the first two, and one season of photometry the third.
- The tests here use public data and released code and apply to any star with a decade-long HARPS-RVBank series.

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and george (Ambikasaran et al. 2015).

Use of generative AI. The author designed the study and made every analysis decision. The analysis code, the literature search and the draft text were produced with the assistance of Claude (Anthropic), a large language model. The results were validated as follows. Every number in the paper is traced, in a provenance table distributed with the code, to the released script, pipeline output or published source that produced it. The adopted Gaussian-process model was implemented twice, in separate code bases (both written with AI assistance), and the two agree within the stated resolution (Appendix A). The false-inclusion computation was checked against nested sampling of the same model and against juliet. The false-alarm, programme-split, alias and forward-model procedures were run on simulated data with

known answers (a Keplerian at the fitted amplitude is recovered in 72% of trials; a signal at the alias period is never mistaken for P_0). Each reference was checked against the publisher's record. The author takes responsibility for the content.

DATA AVAILABILITY

The velocities and indicators used here (Table G1) are available in full as an ancillary file with the arXiv version of this paper and in the deposit below. The frozen data tables with checksums, the outputs of the search of Sect. 2.2, the rejection criteria and decision rules, all nested-sampling outputs, and the code for every figure and number in this paper are deposited at Zenodo (doi:10.5281/zenodo.23249329) and mirrored at <https://github.com/samsonmaximus/hd297396b>.

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TABLE A1
PRIORS.

Parameter	Prior	Range
P (sampling)	Jeffreys	4.20–4.34 d
P (full band, analytic)	Jeffreys	1.05–1000 d
K	uniform	0–30 m s ⁻¹
phase	uniform	0–1
offset, per label	uniform	±50 m s ⁻¹
jitter, per label	log-uniform	0.01–50 m s ⁻¹
β , error scaling	log-uniform	0.1–10
GP amplitude A	log-uniform	0.1–50
GP length scale ℓ	log-uniform	50–5000 d
eccentricity e (1p model)	Beta(0.867, 3.03)	0–1
second period P_2	Jeffreys	150–260 d
outlier fraction π	uniform	0–0.2
outlier variance V	log-uniform	1–10 ⁴ m ² s ⁻²

The eccentricity prior is that of [Kipping \(2013\)](#). The last two rows apply only to the one-outlier mixture of Appendix C.

Sreenivas, K. R., Perdelwitz, V., Tal-Or, L., et al. 2022, A&A, 660, A124, doi: [10.1051/0004-6361/202142612](#)
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APPENDIX

A. PRIORS, SAMPLER AND THE FALSE-INCLUSION PROBABILITY

Table A1 lists the priors.

Resolution of the evidences. Four repeated runs of one noise model with different seeds and live-point counts scatter by 1.71 nats, four times the dynesty internal estimate; changing the implementation moves $\Delta \ln \mathcal{Z}$ by about 1 nat. We treat differences below 2 nats as unresolved.

An independent implementation. We rewrote the adopted model from Eq. (1) in separate code with the same priors. It gives $\Delta \ln \mathcal{Z} = +7.33 \pm 0.35$, $P = 4.26837 \pm 0.00028$ d, $K = 5.18 \pm 0.81$ m s⁻¹ and the same conjunction time to 0.01 d, within the resolution floor and 0.4σ in K of the primary code. In both, the length scale lies at 500–2000 d, far from P_0 .

Full-band evidences and the FIP. With white noise and per-label jitters, the likelihood is Gaussian in the four offsets and in the cosine and sine amplitudes of each circular orbit. We integrate the offsets analytically, and the amplitudes against the prior of Table A1 (uniform in K over 0–30 m s⁻¹ and in phase) through the identity $1/K = \pi^{-1/2} \int_0^\infty s^{-1/2} e^{-sK^2} ds$; an isotropic approximation, used for the two-planet model, changes the evidence by less than 0.01 nats. The period is summed over 1.24×10^5 frequencies, twenty per resolution element, with the Jeffreys prior over 1.05–1000 d. The two-planet evidence conditions the second frequency on the first through a Schur complement, restricted to first frequencies within 10 nats of the one-planet maximum. The four jitters are integrated by importance sampling from a Student- t mixture centred on the zero-, one- and two-planet maxima, with 240 draws and effective sample sizes of 11–46. Two random seeds reproduce the evidences to within 0.5 nats and give FIPs of 1.7 and 2.3×10^{-4} (two planets) and 0.15 and 0.17 (one planet) with all epochs. Restricted to the narrow prior, this computation gives +8.7 and +18.5, within 1 nat of nested sampling of the same model with all epochs. The FIP of a band is $1 - \sum_k p(k|d) \text{TIP}_k$, with TIP_k the posterior probability that the band contains a planet under the k -planet model and equal prior probabilities for $k = 0, 1$ and 2. Essentially all of the one-planet posterior in the band lies within one resolution element of P_0 . A third sinusoid is favoured by 5.0 nats without the discrepant night (not with all epochs), consistent with the slow variability that the white-noise frame does not model.

B. THE NOISE-MODEL LADDER

Table B1 gives the evidence for the planet and its semi-amplitude under six noise models, two extractions and both data sets. On the planet-free model the Matérn process is preferred over white noise by 24.7 nats, yet against a white-noise control in the same code it adds only 1.1 nats of evidence for the planet (+8.87 against +7.77): the correlated noise lives on long timescales and does not compete with the Keplerian.

Our selection rule picks the noise model with the highest planet-free evidence, which is the variant with a seeing-dependent variance term. Its preference is 19.9 nats with the discrepant night and 1.1 nats without, so it encodes one night rather than a property of the noise, and we adopt the plain Matérn process instead, the least favourable model for the planet.

Gaussian-process-weighted false-alarm probabilities. Under the adopted model the velocity amplitude of the process sits at

TABLE B1
MODEL COMPARISON, $\Delta \ln \mathcal{Z}(\text{1p1c} - \text{0p})$, AND THE SEMI-AMPLITUDE K IN THE SAME CELLS.

Noise model	SERVAL		DRS	
	104	103	104	103
$\Delta \ln \mathcal{Z}$, <i>narrow period prior</i>				
White (juliet)	+6.47	+19.71	+6.01	+13.81
Matérn GP (adopted)	+8.87	+27.44	+5.20	+18.13
Matérn + Student- t	+11.72	+26.91	+6.47	...
Quasi-periodic GP	+9.71	...	...	...
Matérn + seeing term	+16.35	+15.50	+12.46	...
Matérn + one-outlier mixture	+25.61	+22.00	+13.95	...
K (m s^{-1})				
Matérn GP (adopted)	5.52 ± 0.84	5.59 ± 0.37	5.72 ± 1.03	5.54 ± 0.64
Quasi-periodic GP (ML)	5.63 ± 0.99	5.47 ± 0.83	...	...
Matérn + Student- t	5.64 ± 0.85	5.61 ± 0.36	5.30 ± 1.18	...
Matérn + seeing term	5.27 ± 0.43	5.39 ± 0.58	...	...
Matérn + one-outlier mixture	5.57 ± 0.37	5.54 ± 0.37	5.56 ± 0.65	...
<i>Other comparisons, adopted model, SERVAL, 104 epochs</i>				
Full search band, Jeffreys 1.05–1000 d				+3.53
P_0 against the 1.301-d alias, matched priors				+6.02
Second signal at 200.9 d, 2p1c2c–1p1c				+5.98

Positive values favour the first model. The 103-epoch columns omit the discrepant night. Typical dynesty uncertainties are 0.3–0.5 nats; repeated runs scatter by up to 2 nats. The white-noise row uses juliet (Espinoza et al. 2019). The quasi-periodic K is the generalised-least-squares amplitude at the maximum-likelihood covariance. Entries marked ... were not run.

the lower edge of its prior at maximum likelihood; its posterior spreads over a few m s^{-1} on 500–2000 d (median 3 m s^{-1} , 95% upper limit 8 m s^{-1} under a uniform amplitude prior). Weighting the periodogram by the maximum-likelihood covariance gives $\Delta\chi^2 = 32.4$ and 44.0 at P_0 and 12 exceedances in 10 000 draws of the same covariance, a FAP of 1.2×10^{-3} . At the medians of the nested-sampling posterior (5 m s^{-1} on 1200 d) or its upper range (9 m s^{-1} on 750 d) the values are 42 and 39 for all epochs, still the highest peak, with no exceedance in 3000 draws. The quasi-periodic process gives $\Delta\chi^2 = 32.2$ and 43.9 , with 4 and 0 exceedances in 5000 draws.

The two extractions. Under white noise the extractions agree on the evidence to 0.46 nats. Under the adopted model they differ by 3.7 nats, about twice the run-to-run scatter, because the DRS needs larger jitters in the post-upgrade and minor-programme labels (4.6 and 5.2 m s^{-1} against 3.5 and 4.2 m s^{-1} for SERVAL), with which the Keplerian competes.

C. AN AT-MOST-ONE-OUTLIER LIKELIHOOD

A multivariate Student- t likelihood has one latent scale for the whole series, and a per-point Gaussian mixture sums over 2^N assignments and is not tractable under an exact Gaussian process. The mixture with at most one discrepant epoch of unknown identity has $N + 1$ terms,

$$p(\mathbf{r}) = (1 - \pi) \mathcal{N}(\mathbf{r}; \mathbf{0}, \Sigma) + \frac{\pi}{N} \sum_{i=1}^N \mathcal{N}(\mathbf{r}; \mathbf{0}, \Sigma + V \mathbf{e}_i \mathbf{e}_i^T), \quad (\text{C1})$$

each a rank-1 update of the same covariance. With $\mathbf{a} = \Sigma^{-1} \mathbf{r}$ and $\mathbf{b} = \text{diag}(\Sigma^{-1})$, the matrix determinant lemma and the Sherman–Morrison identity give $\ln |\Sigma + V \mathbf{e}_i \mathbf{e}_i^T| = \ln |\Sigma| + \ln(1 + V b_i)$ and $\mathbf{r}^T (\Sigma + V \mathbf{e}_i \mathbf{e}_i^T)^{-1} \mathbf{r} = \mathbf{r}^T \Sigma^{-1} \mathbf{r} - V a_i^2 / (1 + V b_i)$, so all N terms cost $O(N)$ after the Cholesky factorisation.

Told nothing about which epoch to examine, the model assigns BJD 2454922.53 a posterior probability of 1.000 in both extractions, with the runner-up at 2×10^{-26} . On the 103-epoch set the probability that any epoch is discrepant falls to 0.40, no single epoch exceeds 0.022, and a two-outlier variant loses by 13 nats. Applied to the two HARPS-RVBank planets of Sreenivas et al. (2022), it recovers both, with periods and semi-amplitudes within 2% of the published values, and finds no outlier on either star.

D. THE DISCREPANT NIGHT

A Gaussian likelihood must inflate the noise of all 47 epochs of programme 072.C-0488 to accommodate one excursion: in the white-noise frame the fitted jitter is 7.7 m s^{-1} with the night and 3.1 m s^{-1} without it. Without any provision for the night, that epoch supplies 34% of the $\Delta\chi^2$ at P_0 and ranks first of 104; once the seeing-keyed term down-weights it, it supplies 0.6% and ranks 47th, and no epoch exceeds 12%. In the whitened residuals of the adopted model it is a 6.0σ point; without it the largest $|z|$ is 2.46, close to the 2.7 expected for the largest of 103 unit Gaussian draws. Two independent devices, one keyed to the archival seeing record and one to the residuals alone, pick the same night. We keep it in the primary data set because it passes every rejection criterion.

E. ASAS-SN PHOTOMETRY

The ASAS-SN DR9 aperture photometry shows an 11.9 ± 3.1 mmag modulation at P_0 . The star, at $g = 10.76$, is about 0.7 mag brighter than the ASAS-SN saturation limit (Winecki & Kochanek 2024). The fitted amplitude runs from 5.7 to 12.1 mmag as the clipping threshold changes, and the quoted significance uses per-epoch errors about half the measured scatter. TESS, which does not saturate on this star, excludes the amplitude by more than a factor of 60 in every sector and 170 in the stitched light curve. The saturation-corrected reduction of Winecki & Kochanek (2024), with 199 V-band and 1482 g-band epochs over 8.1 yr, returns 1.9 and 1.5 mmag at P_0 (3σ upper limits 7.3 and 5.5 mmag), and would have recovered the aperture signal at 93% of its

TABLE G1
FIRST ROWS OF THE NIGHTLY VELOCITIES.

BJD _{TDB} −2450000	RV (m s ^{−1})	σ_{RV} (m s ^{−1})	ΔLW (m ² s ^{−2})	$\sigma_{\Delta LW}$ (m ² s ^{−2})	Label
3031.78326	+5.54	1.38	−14.7	2.9	pre_072
3034.82031	−0.69	1.24	−8.3	1.8	pre_072
3358.80623	−9.12	1.40	−8.3	2.0	pre_oth
3359.83542	+0.08	1.38	−14.8	1.7	pre_oth
3360.79010	+4.35	1.42	−7.3	1.7	pre_oth
3363.76791	+1.52	1.50	−2.3	1.8	pre_oth
3484.59038	+10.50	1.11	−5.2	1.2	pre_072
3491.53553	+6.23	1.54	−2.5	2.5	pre_072
...	...	...	...	...	...

SERVAL velocities corrected for the nightly zero point, binned by night and label. The full table (104 epochs, with DRS velocities and all nine indicators) accompanies the arXiv version of this paper.

amplitude. After per-camera offsets and a 3σ clip, 1376 g epochs over 8.0 yr bound any modulation in 25–40 d below 8 mmag at 90% recovery (40–50 d was not searched); the strongest feature in the band, 5.7 mmag at 26.83 d, fails a block bootstrap.

F. TESS PHOTOMETRY

TESS observed HD 297396 in sectors 8 and 9 (2019), 35 and 36 (2021), 62 and 63 (2023) and 89 and 90 (2025). We use SPOC PDCSAP photometry (Jenkins et al. 2016) where it exists and QLP full-frame-image photometry (Huang et al. 2020a,b) otherwise, keeping points with a zero quality flag: 68 461 points over 2257 d. Each sector is normalised by its median and detrended with a $5P_0$ running median, which retains 96% of a signal at P_0 . The null distribution is built per sector by fitting at 10^3 random trial periods and by fitting at P_0 after 10^3 circular shifts of the flux; the two agree to 10%. The stitched light curve gives 28 ppm at P_0 against a null of 48 ppm. A 30–50-d modulation is longer than a sector and is removed by any detrending, so TESS cannot measure the rotation period. A box-least-squares search finds no transit at the candidate period.

G. RADIAL VELOCITIES

Table G1 shows the format of the table of velocities and indicators that accompanies this paper.